PARIS PANTHÉON-ASSAS UNIVERSITÉ

EFREI PARIS

Project - Database Systems
Quantitative Finance

—Big Data & Machine Learning—

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Contents

1 Introduction

In the context of the Database Systems course (INGE1-APP-BDML), this project focuses on the design and implementation of a relational database applied to investment portfolio management and quantitative finance.
The objective is to model the operations of a financial firm managing its clients' assets: tracking financial instruments, price history, transaction recording, and portfolio performance computation.
The project is structured around three main components: database modeling (CDM/LDM), SQL implementation under MySQL with integrity constraints and advanced queries, as well as the development of a Python console application enabling interaction with the data and display of financial statistics.

Software repository: https://github.com/bm1nhtr/QuFiSQL

2 Domain Description

2.1 Domain Overview

The database models an investment portfolio management system, as found within asset management firms or quantitative investment funds. As such, the system tracks clients' positions on financial markets, values their portfolios, and maintains a history of transactions and closing prices.

A client is an investor (individual or institutional) characterized by their Assets Under Management (AUM) and risk profile (*conservative*, *balanced*, *dynamic*, *aggressive*).
Each client is monitored by a portfolio manager, an employee of the firm, who holds a specialization (*equities*, *fixed income*, *derivatives*, *multi-asset*). A manager oversees multiple clients, but a client is assigned to a single manager at any given time.

Each client holds one or more portfolios. A portfolio has a reference currency, an investment strategy (*value*, *growth*, *momentum*, *market-neutral*, *index-tracking*), a creation date, and a net asset value recalculated periodically.
A portfolio is a wrapper that contains positions across various financial instruments.

A financial instrument is a tradable asset: equity, bond, ETF, or derivative. It is identified by a unique ticker, linked to an economic sector and a quotation currency, and characterized by its annualized volatility.
Each instrument has an associated series of historical prices: for each trading date, the closing price, traded volume, and daily return are recorded, enabling quantitative calculations (volatility, correlation, performance).

The core of the model is the position: it represents the holding of a given instrument within a given portfolio at a given valuation date.
A position is therefore a ternary association (`Portfolio` × `Instrument` × `Date`) carrying attributes: the quantity held, the average acquisition price, the weight of the line in the portfolio (between 0 and 1), and the *latent PnL* (unrealized *Profit and Loss*).
The sum of the weights of all positions in a portfolio at a given date must not exceed 100%.

Finally, each buy or sell movement is recorded as a transaction: it carries a direction (*buy* or *sell*), a quantity, an execution price, a date, and brokerage fees.

A transaction is linked to a specific portfolio and concerns a specific instrument. The transaction history allows positions to be reconstructed and realized performance to be computed.
This scope, rich in quantitative management rules, provides a solid and realistic foundation for the relational modeling required.

2.2 Business Rules

The rules (see Table 1), oriented toward quantitative finance, govern the data and guide the design of SQL constraints (`NOT NULL`, `UNIQUE`, `CHECK`, `FK`).

Code	**Business Rule**
RM01	A client is monitored by exactly one manager at any given time; a manager oversees 0 to N clients.
RM02	A client holds at least one portfolio; each portfolio belongs to a single client.
RM03	A client's risk profile must be one of: *conservative*, *balanced*, *dynamic*, *aggressive*.
RM04	A client's *AUM* must be positive; an aggressive profile requires an AUM $\geq$ 100,000 € (regulatory consistency).
RM05	A position links a portfolio, an instrument, and a valuation date (ternary association).
RM06	The weight of a position is between 0 and 1 (0% to 100%).
RM07	The sum of the weights of all positions in a given portfolio at a given date must not exceed 1 (100%).
RM08	The quantity of a position is strictly positive; the average price is strictly positive.
RM09	The *latent PnL* of a position may be positive or negative (unrealized gain or loss).
RM10	An instrument has a unique ticker and a type among: *equity*, *bond*, *ETF*, *derivative*.
RM11	The volatility of an instrument is non-negative (expressed as an annualized percentage).
RM12	A transaction has a direction of either *BUY* or *SELL*; its quantity and execution price are strictly positive.
RM13	Transaction fees are non-negative and must not exceed 5% of the gross amount.
RM14	A transaction concerns exactly one portfolio and involves exactly one instrument.
RM15	For a given instrument, at most one historical price exists per trading date (uniqueness of ticker + date).
RM16	The closing price of a historical price entry is strictly positive; volume is non-negative.
RM17	The currency of a portfolio and an instrument follows the ISO 4217 format (3 uppercase letters, e.g. `EUR`, `USD`).
RM18	The valuation date of a position cannot be earlier than the creation date of the portfolio.

Table 1: System business rules

2.3 Data Dictionary

Rather than treating the database as a simple set of relational tables, an institutional asset manager interprets it as a temporal map of the investment lifecycle. The structure models the transfer of capital from client mandates through to active risk-taking, recording execution quality, portfolio valuation, and performance generation[1]. The different layers of the database are designed to support the distinct components of institutional asset management (see Fig 1).

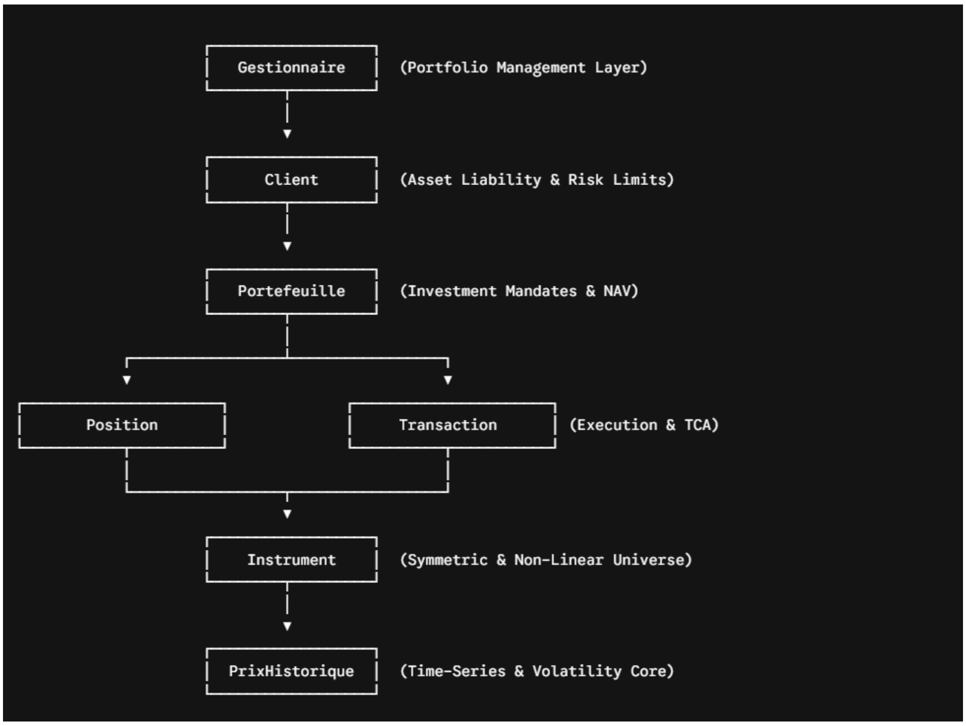

Figure 1: Institutional view of the database schema

The `Gestionnaire` and `Client` tables model the operational hierarchy of the asset management firm and formalise the investment parameters specific to each client. The risk profile classification (`profil_risque`) and patrimonial constraints — in particular the minimum assets under management threshold required for aggressive strategies — programmatically prevent any breach of investment suitability rules.

The `Portefeuille` table represents distinct investment strategies, each operating in a specific reference currency (`EUR` or `USD`). Daily updates to the net asset value (`valeur_liquidative`) provide the common denominator for computing portfolio weights and risk-return metrics.

The `Position` table serves as a daily point-in-time holding ledger. It tracks the quantity held, the average cost price, the portfolio weight, and the unrealised profit and loss (`pnl_latent`). It constitutes the primary dataset for computing systematic factor exposures, tracking error, and idiosyncratic risk contributions.

The `Transaction` table records every buy or sell order executed, capturing quantity, execution price, and direct transaction costs. Comparing this register against historical market prices is essential for Transaction Cost Analysis (TCA) and market impact modelling.

The `Instrument` and `PrixHistorique` tables provide asset reference data and historical price series. This layer supports the computation of daily returns (`rendement_jour`), price and volatility trend analysis, and input parameters for option pricing models.

[1] Hedgeco

The data dictionary details the 7 tables of the model. The `Position` table originates from the ternary association and has a composite primary key (`id_portefeuille`, `id_instrument`, `date_valo`).

Attribute	**SQL Type**	**Constraints**	**Description**
id_client	INT	PK, AUTO_INCREMENT	Unique client identifier
nom	VARCHAR(60)	NOT NULL	Client surname
prenom	VARCHAR(60)	NOT NULL	Client first name
email	VARCHAR(120)	UNIQUE, NOT NULL	Contact email
aum	DECIMAL(15,2)	NOT NULL, CHECK > 0	Assets under management in € (RM04)
profil_risque	ENUM('prudent', 'équilibré', 'dynamique', 'agressif')	NOT NULL	Risk profile (RM03)
date_entree	DATE	NOT NULL	Client onboarding date
#id_gestionnaire	INT	FK to Gestionnaire, NOT NULL	Assigned portfolio manager (RM01)

Table 2: Data Dictionary — Table `Client`

Attribute	**SQL Type**	**Constraints**	**Description**
id_gestionnaire	INT	PK, AUTO_INCREMENT	Unique portfolio manager identifier
nom	VARCHAR(60)	NOT NULL	Manager surname
prenom	VARCHAR(60)	NOT NULL	First name
email	VARCHAR(120)	UNIQUE, NOT NULL	Professional email
date_embauche	DATE	NOT NULL	Hiring date
specialite	VARCHAR(100)	NULL	Manager's area of specialisation

Table 3: Data Dictionary — Table `Gestionnaire`

Attribute	**SQL Type**	**Constraints**	**Description**
id_instrument	INT	PK, AUTO_INCREMENT	Unique instrument identifier
ticker	VARCHAR(12)	UNIQUE, NOT NULL	Exchange ticker symbol (RM10)
nom	VARCHAR(120)	NOT NULL	Instrument name
type_instrument	ENUM('ACTION', 'OBLIGATION', 'ETF', 'DERIVE')	NOT NULL	Instrument type (RM10)
secteur	VARCHAR(100)	NULL	Economic sector

Attribute	SQL Type	Constraints	Description
devise	CHAR(3)	NOT NULL, CHECK ISO format	Quotation currency (RM17)
volatilite	DECIMAL(10,4)	CHECK $\geq$ 0, DEFAULT 0	Annualised volatility (RM11)

Table 4: Data Dictionary — Table `Instrument`

Attribute	SQL Type	Constraints	Description
id_portefeuille	INT	PK, AUTO_INCREMENT	Unique portfolio identifier
nom	VARCHAR(100)	NOT NULL	Portfolio name
devise_base	CHAR(3)	NOT NULL, CHECK ISO format	Reference currency (RM17)
strategie	ENUM('VALUE', 'GROWTH', 'MOMENTUM', 'NEUTRAL', 'INDICIEL')	NOT NULL	Investment strategy
date_creation	DATE	NOT NULL	Creation date
valeur_liquidative	DECIMAL(18,2)	NOT NULL, DEFAULT 0, CHECK $\geq$ 0	Portfolio net asset value
#id_client	INT	FK to Client, NOT NULL	Owning client (RM02)

Table 5: Data Dictionary — Table `Portefeuille`

Attribute	SQL Type	Constraints	Description
#id_portefeuille	INT	PK part 1, FK to Portefeuille	Portfolio concerned (RM05)
#id_instrument	INT	PK part 2, FK to Instrument	Instrument held (RM05)
date_valo	DATE	PK part 3	Valuation date (RM05, RM18)
quantite	DECIMAL(15,4)	NOT NULL, CHECK $>$ 0	Quantity held (RM08)
prix_moyen	DECIMAL(15,4)	NOT NULL, CHECK $>$ 0	Average acquisition price (RM08)
poids	DECIMAL(5,4)	NOT NULL, CHECK BETWEEN 0 AND 1	Portfolio weight (RM06, RM07)
pnl_latent	DECIMAL(18,2)	NOT NULL, DEFAULT 0	Unrealised PnL (RM09)

Table 6: Data Dictionary — Table `Position` (derived from the ternary association)

Attribute	SQL Type	Constraints	Description
id_prix	INT	PK, AUTO_INCREMENT	Unique price record identifier
#id_instrument	INT	FK to Instrument, NOT NULL	Quoted instrument (RM15)
date_cotation	DATE	NOT NULL	Quotation date (RM15)
cours_cloture	DECIMAL(15,4)	NOT NULL, CHECK > 0	Closing price (RM16)
volume	BIGINT	CHECK $\geq$ 0, DEFAULT 0	Traded volume (RM16)
rendement_jour	DECIMAL(10,6)	NULL	Daily return (RM16)

Table 7: Data Dictionary — Table `PrixHistorique`

Attribute	SQL Type	Constraints	Description
id_transaction	INT	PK, AUTO_INCREMENT	Unique transaction identifier
sens	ENUM('ACHAT', 'VENTE')	NOT NULL	Order direction (RM12)
quantite	DECIMAL(15,4)	NOT NULL, CHECK > 0	Traded quantity (RM12)
prix_execution	DECIMAL(15,4)	NOT NULL, CHECK > 0	Execution price (RM12)
date_transaction	DATETIME	NOT NULL	Operation timestamp
frais	DECIMAL(18,2)	NOT NULL, CHECK ≥ 0	Brokerage fees (RM13)
#id_portefeuille	INT	FK to Portefeuille, NOT NULL	Portfolio concerned (RM14)
#id_instrument	INT	FK to Instrument, NOT NULL	Instrument traded (RM14)

Table 8: Data Dictionary — Table `Transaction`

3 Database Modeling

3.1 Conceptual Data Model (CDM)

The CDM below, using Merise notation, consists of 7 entities. The core of the model is the ternary association POSITION, which links Portfolio, Instrument, and the temporal dimension (valuation date), and carries the attributes weight, latent_PnL, quantity, and average_price.

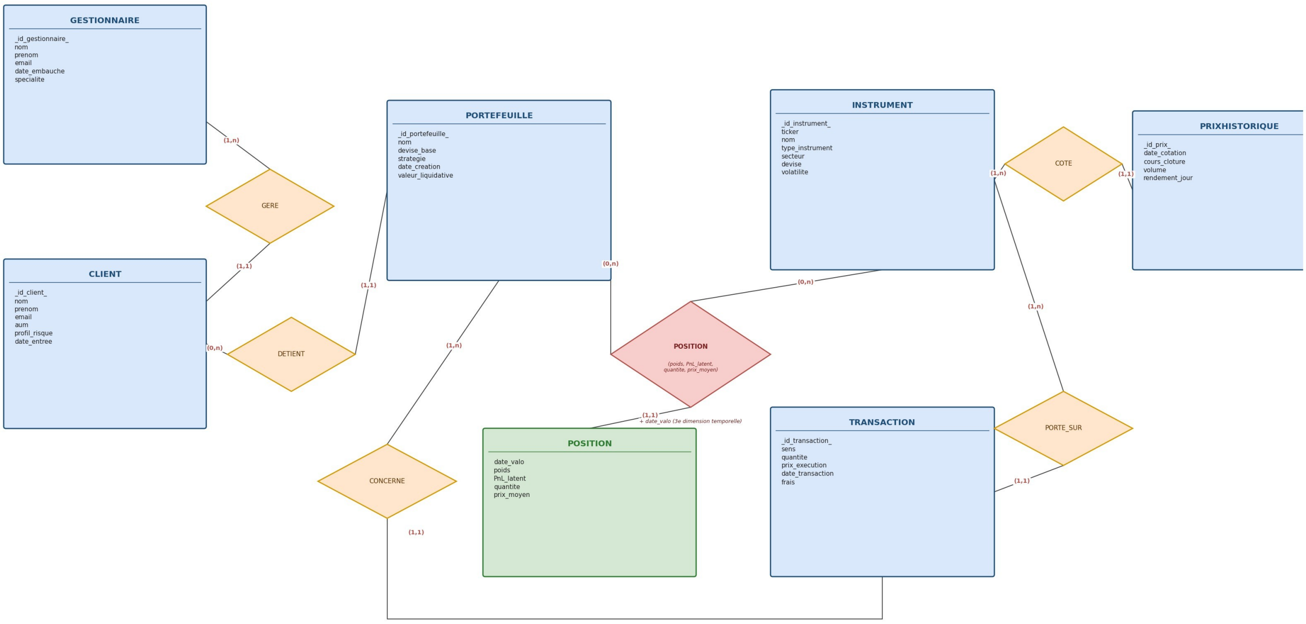

Figure 2: CDM Schema

Project Requirement	Status
At least 5 distinct entities	7 entities: `Gestionnaire`, `Client`, `Portefeuille`, `Instrument`, `Position`, `Transaction`, `PrixHistorique`
18 quantitative business rules, CHECK constraints, and composite keys	Constraints implemented in the relational model and business rules
At least 1 ternary OR attribute-carrying association	`POSITION`: ternary association (`Portefeuille` × `Instrument` × `Date`) carrying attributes
Model in 3rd Normal Form	No transitive dependencies (see justification below)
Sufficiently rich domain	The quantitative finance domain allows manipulation of multiple entities, constraints, and complex relationships

Table 9: Verification of Project Requirements

Mandatory Requirements (see Table 9)

- **1NF:** All attributes are atomic. The price history is broken down into rows of `PrixHistorique` (one row per date), rather than repeated columns.
- **2NF:** The only table with a composite key is `Position`. Its attributes (`quantite`, `prix_moyen`, `poids`, `pnl_latent`) depend on the full triplet (`id_portefeuille`, `id_instrument`, `date_valo`) and not on any single part of it.
- **3NF:** No transitive dependencies. The `secteur` and `volatilite` fields are stored in `Instrument` and are not duplicated in `Position` or `Transaction`. The `profil_risque`

field is stored in `Client` and is not copied into `Portefeuille`. Each non-key attribute depends solely on the primary key of its own table.

3.2 Logical Data Model (LDM)

The LDM is derived by applying the CDM → LDM transformation rules:

- **Entities:** Each entity becomes a table; its identifier becomes the primary key.
- $(x, 1) - (x, n)$ **associations:** Translated as a foreign key on the $(x, 1)$ side: `GERE`, `DETIENT`, `CONCERNE`, `PORTE_SUR`, `COTE`.
- **Attribute-carrying ternary association:** `POSITION` becomes a table whose primary key is composed of the three identifiers (`id_portefeuille`, `id_instrument`, `date_valo`); its carried attributes (`poids`, `pnl_latent`, `quantite`, `prix_moyen`) become columns of this table.

3.3 Textual Notation of Tables

Convention: Primary key underlined, foreign keys prefixed with #.

- <u>`Gestionnaire`</u>
  (<u>`id_gestionnaire`</u>, `nom`, `prenom`, `email`, `date_embauche`, `specialite`)
- <u>`Client`</u>
  (<u>`id_client`</u>, `nom`, `prenom`, `email`, `aum`, `profil_risque`, `date_entree`, #`id_gestionnaire`)
- <u>`Portefeuille`</u>
  (<u>`id_portefeuille`</u>, `nom`, `devise_base`, `strategie`, `date_creation`, `valeur_liquidative`, #`id_client`)
- <u>`Instrument`</u>
  (<u>`id_instrument`</u>, `ticker`, `nom`, `type_instrument`, `secteur`, `devise`, `volatilite`)
- <u>`Position`</u>
  (#`id_portefeuille`, #`id_instrument`, <u>`date_valo`</u>, `quantite`, `prix_moyen`, `poids`, `pnl_latent`)
- <u>`Transaction`</u>
  (<u>`id_transaction`</u>, `sens`, `quantite`, `prix_execution`, `date_transaction`, `frais`, #`id_portefeuille`, #`id_instrument`)
- <u>`PrixHistorique`</u>
  (<u>`id_prix`</u>, `date_cotation`, `cours_cloture`, `volume`, `rendement_jour`, #`id_instrument`)

3.4 Summary of Foreign Keys

Table	Foreign Key	Reference	Origin (CDM association)
Client	#id_gestionnaire	Gestionnaire(id_gestionnaire)	GERE
Portefeuille	#id_client	Client(id_client)	DETIENT
Position	#id_portefeuille	Portefeuille(id_portefeuille)	POSITION (ternary)
Position	#id_instrument	Instrument(id_instrument)	POSITION (ternary)
Transaction	#id_portefeuille	Portefeuille(id_portefeuille)	CONCERNE
Transaction	#id_instrument	Instrument(id_instrument)	PORTE_SUR

Table	Foreign Key	Reference	Origin (CDM association)

Table 10: Summary of Foreign Keys

4 SQL Database

4.1 Creation Script (DDL)

The complete DDL script was written in MySQL and structured in three distinct phases: database initialization, dropping existing tables, and creating tables with their integrity constraints.

Database Initialization

The first phase creates the database specifying the `utf8mb4` encoding to ensure full Unicode character support (including French characters), then drops existing tables in reverse dependency order to allow re-execution without errors (see Fig 3).

```sql
-- ============================================================
--  Quant Finance Database – Script de création MySQL
-- ============================================================

CREATE DATABASE IF NOT EXISTS quant_finance
    CHARACTER SET utf8mb4        -- Encodage UTF-8 complet (4 octets), supporte tous les caractères Unicode
    COLLATE utf8mb4_unicode_ci;  -- Interclassement insensible à la casse, conforme Unicode
                                 -- (nécessaire car les données contiennent des caractères français : é, è, ê, ç...)

USE quant_finance;

-- ============================================================
--  DROP (ordre inverse des dépendances FK)
-- ============================================================
DROP TABLE IF EXISTS PrixHistorique;
DROP TABLE IF EXISTS Transaction;
DROP TABLE IF EXISTS Position;
DROP TABLE IF EXISTS Instrument;
DROP TABLE IF EXISTS Portefeuille;
DROP TABLE IF EXISTS Client;
DROP TABLE IF EXISTS Gestionnaire;
```

Figure 3: Database initialization and dropping of existing tables

Core Table Creation

The `Gestionnaire` and `Client` tables form the backbone of the relational model. The `Client` table notably includes a composite `CHECK` constraint verifying that any client with an *aggressive* profile holds an AUM greater than or equal to 100 000 € (business rule RM04) (see Fig 4).
The `Portefeuille` and `Instrument` tables complete the base model. The currency field is validated by a regular expression enforcing the ISO 4217 format (three uppercase letters), in accordance with business rule RM17 (see Fig 5).

Transactional Table Creation

The `Position` table relies on a composite primary key (`id_portefeuille`, `id_instrument`, `date_valo`), thereby modeling the ternary association defined in the conceptual data model (business rule RM05) (see Fig 6).

```sql
-- ============================================================
-- 1. Gestionnaire
-- ============================================================
CREATE TABLE Gestionnaire (
    id_gestionnaire INT          NOT NULL AUTO_INCREMENT,
    nom             VARCHAR(100) NOT NULL,
    prenom          VARCHAR(100) NOT NULL,
    email           VARCHAR(150) NOT NULL,
    date_embauche   DATE         NOT NULL,
    specialite      VARCHAR(100),

    CONSTRAINT pk_gestionnaire PRIMARY KEY (id_gestionnaire),
    CONSTRAINT uq_gestionnaire_email UNIQUE (email),
    CONSTRAINT chk_gestionnaire_email CHECK (email LIKE '%@%.%')
);

-- ============================================================
-- 2. Client
-- ============================================================
CREATE TABLE Client (
    id_client       INT           NOT NULL AUTO_INCREMENT,
    nom             VARCHAR(100)  NOT NULL,
    prenom          VARCHAR(100)  NOT NULL,
    email           VARCHAR(150)  NOT NULL,
    aum             DECIMAL(18,2) NOT NULL,
    profil_risque   ENUM('prudent','équilibré','dynamique','agressif') NOT NULL,
    date_entree     DATE          NOT NULL,
    id_gestionnaire INT           NOT NULL,

    CONSTRAINT pk_client         PRIMARY KEY (id_client),
    CONSTRAINT uq_client_email   UNIQUE (email),
    CONSTRAINT fk_client_gest    FOREIGN KEY (id_gestionnaire)
                                 REFERENCES Gestionnaire(id_gestionnaire)
                                 ON DELETE RESTRICT
                                 ON UPDATE CASCADE,
    CONSTRAINT chk_client_aum    CHECK (aum > 0),
    CONSTRAINT chk_client_aum_agressif CHECK (
        profil_risque <> 'agressif' OR aum >= 100000
    ),
    CONSTRAINT chk_client_email  CHECK (email LIKE '%@%.%')
);
```

Figure 4: Creation of the `Gestionnaire` and `Client` tables

The `PrixHistorique` table enforces the uniqueness of the (`id_instrument`, `date_cotation`) pair via a `UNIQUE` constraint, in accordance with rule RM15 (see Fig 7).
Finally, the `Transaction` table encodes the fee constraint — fees may not exceed 5 % of the gross amount — directly within a `CHECK` clause (business rule RM13) (see Fig 8).

4.2 Dataset (DML)

The DML script populates all tables in foreign key dependency order, ensuring business consistency of the inserted data. Table 11 summarizes the volume of data generated.

> **Note:** All inserted data is entirely **fictitious and simulated**. It was generated solely for the purpose of testing and validating queries. Any resemblance to real financial data (prices, volumes, transactions) is purely coincidental.

Table	Rows	Remarks
`Gestionnaire`	6	6 distinct specializations (equities, bonds, derivatives, ETFs...)

Table	Rows	Remarks
`Client`	8	4 risk profiles represented; AUM consistent with business rules
`Portefeuille`	8	EUR and USD currencies; varied strategies (conservative, balanced, aggressive)
`Instrument`	15	6 equities + 3 bonds + 3 ETFs + 3 derivatives
`PrixHistorique`	210	Time series from August to December 2024 across all 15 instruments
`Position`	35	Weightings consistent with the risk profile of each portfolio
`Transaction`	55	Chronological order respected: every `BUY` precedes the corresponding `SELL`

Table 11: Volume of data inserted per table

The figures below present a representative extract (< first 15 rows) of each table for illustration purposes.

Reference Tables

```sql
INSERT INTO Gestionnaire (nom, prenom, email, date_embauche, specialite) VALUES
('Dupont',    'Alexandre', 'a.dupont@quantfin.fr',    '2018-03-15', 'Actions européennes'),
('Martin',    'Sophie',    's.martin@quantfin.fr',    '2019-07-01', 'Obligations souveraines'),
('Bernard',   'Julien',    'j.bernard@quantfin.fr',   '2020-01-20', 'Dérivés et produits structurés'),
```

Figure 9: Extract from the `Gestionnaire` table

```sql
INSERT INTO Client (nom, prenom, email, aum, profil_risque, date_entree, id_gestionnaire) VALUES
('Fontaine',  'Pierre',    'p.fontaine@gmail.com',    45000.00,   'prudent',    '2020-02-10', 2),
('Rousseau',  'Marie',     'm.rousseau@gmail.com',    78000.00,   'équilibré',  '2019-06-15', 1),
('Leclerc',   'Antoine',   'a.leclerc@gmail.com',     120000.00,  'agressif',   '2021-01-08', 3),
```

Figure 10: Extract from the `Client` table

```sql
INSERT INTO Portefeuille (nom, devise_base, strategie, date_creation, valeur_liquidative, id_client) VALUES
('PF Sécurité Fontaine',     'EUR', 'Obligations court terme',         '2020-02-15', 46200.00,  1),
('PF Équilibré Rousseau',    'EUR', 'Mix 60% actions / 40% obli',      '2019-06-20', 82500.00,  2),
('PF Agressif Leclerc',      'EUR', 'Actions croissance + dérivés',    '2021-01-10', 135000.00, 3),
```

Figure 11: Extract from the `Portefeuille` table

```sql
INSERT INTO Instrument (ticker, nom, type_instrument, secteur, devise, volatilite) VALUES
-- Actions (6)
('MC.PA',   'LVMH Moët Hennessy',          'action',    'Luxe',           'EUR', 0.2215),
('TTE.PA',  'TotalEnergies SE',             'action',    'Énergie',        'EUR', 0.1987),
('SAN.PA',  'Sanofi SA',                    'action',    'Santé',          'EUR', 0.1654),
('AIR.PA',  'Airbus SE',                    'action',    'Aéronautique',   'EUR', 0.2401),
('AAPL',    'Apple Inc.',                   'action',    'Technologie',    'USD', 0.2632),
('MSFT',    'Microsoft Corporation',        'action',    'Technologie',    'USD', 0.2189),
-- Obligations (3)
('OAT10',   'OAT France 10 ans 3.0%',       'obligation','Souverain',      'EUR', 0.0412),
('BUND10',  'Bund Allemagne 10 ans 2.5%',   'obligation','Souverain',      'EUR', 0.0389),
('CORP-SAN','Obligation Sanofi 5 ans 2.8%', 'obligation','Santé',          'EUR', 0.0521),
-- ETF (3)
('CAC40',   'Amundi ETF CAC 40',            'ETF',       'Indice FR',      'EUR', 0.1823),
('SP500',   'iShares Core S&P 500 ETF',     'ETF',       'Indice US',      'USD', 0.1654),
('MSCI-EM', 'Xtrackers MSCI Emerging',      'ETF',       'Marchés émerg.', 'USD', 0.2210),
-- Dérivés (3)
('CAC-FUT', 'Future CAC 40 Déc. 2024',      'dérivé',    'Indice FR',      'EUR', 0.2954),
('AAPL-C',  'Call AAPL Strike 200 Jan25',   'dérivé',    'Technologie',    'USD', 0.4512),
('EUR-PUT', 'Put EUR/USD Strike 1.05 Mar25','dérivé',    'Forex',          'USD', 0.3187);
```

Figure 12: Extract from the `Instrument` table

Transactional Tables

The `PrixHistorique` table forms the analytical core of the dataset: it covers a five-month time series (August–December 2024) for all 15 instruments, totaling approximately 210 records that enable the computation of daily returns and historical volatilities.

```sql
-- ==============================================================
--  3. Portefeuille
-- ==============================================================
CREATE TABLE Portefeuille (
    id_portefeuille   INT            NOT NULL AUTO_INCREMENT,
    nom               VARCHAR(150)   NOT NULL,
    devise_base       CHAR(3)        NOT NULL,
    strategie         VARCHAR(100),
    date_creation     DATE           NOT NULL,
    valeur_liquidative DECIMAL(18,2) NOT NULL DEFAULT 0,
    id_client         INT            NOT NULL,

    CONSTRAINT pk_portefeuille     PRIMARY KEY (id_portefeuille),
    CONSTRAINT fk_portef_client    FOREIGN KEY (id_client)
                                   REFERENCES Client(id_client)
                                   ON DELETE CASCADE
                                   ON UPDATE CASCADE,
    CONSTRAINT chk_portef_devise   CHECK (devise_base REGEXP '^[A-Z]{3}$'),
    CONSTRAINT chk_portef_vl       CHECK (valeur_liquidative >= 0)
);

-- ==============================================================
--  4. Instrument
-- ==============================================================
CREATE TABLE Instrument (
    id_instrument   INT            NOT NULL AUTO_INCREMENT,
    ticker          VARCHAR(20)    NOT NULL,
    nom             VARCHAR(150)   NOT NULL,
    type_instrument ENUM('action','obligation','ETF','dérivé') NOT NULL,
    secteur         VARCHAR(100),
    devise          CHAR(3)        NOT NULL,
    volatilite      DECIMAL(10,4)           DEFAULT 0,

    CONSTRAINT pk_instrument       PRIMARY KEY (id_instrument),
    CONSTRAINT uq_instrument_ticker UNIQUE (ticker),
    CONSTRAINT chk_instrument_devise    CHECK (devise REGEXP '^[A-Z]{3}$'),
    CONSTRAINT chk_instrument_volatilite CHECK (volatilite >= 0)
);
```

Figure 5: Creation of the `Portefeuille` and `Instrument` tables

```sql
-- ==============================================================
--  5. Position  (clé primaire composite)
-- ==============================================================
CREATE TABLE Position (
    id_portefeuille INT           NOT NULL,
    id_instrument   INT           NOT NULL,
    date_valo       DATE          NOT NULL,
    quantite        DECIMAL(18,6) NOT NULL,
    prix_moyen      DECIMAL(18,4) NOT NULL,
    poids           DECIMAL(5,4)  NOT NULL DEFAULT 0,
    pnl_latent      DECIMAL(18,2) NOT NULL DEFAULT 0,

    CONSTRAINT pk_position PRIMARY KEY (id_portefeuille, id_instrument, date_valo),
    CONSTRAINT fk_pos_portef  FOREIGN KEY (id_portefeuille)
                              REFERENCES Portefeuille(id_portefeuille)
                              ON DELETE CASCADE
                              ON UPDATE CASCADE,
    CONSTRAINT fk_pos_instrum FOREIGN KEY (id_instrument)
                              REFERENCES Instrument(id_instrument)
                              ON DELETE RESTRICT
                              ON UPDATE CASCADE,
    CONSTRAINT chk_pos_quantite    CHECK (quantite > 0),
    CONSTRAINT chk_pos_prix_moyen  CHECK (prix_moyen > 0),
    CONSTRAINT chk_pos_poids       CHECK (poids BETWEEN 0 AND 1)
);
```

Figure 6: Creation of the `Position` table

```sql
-- ==============================================================
--  6. PrixHistorique
-- ==============================================================
CREATE TABLE PrixHistorique (
    id_prix         INT             NOT NULL AUTO_INCREMENT,
    date_cotation   DATE            NOT NULL,
    cours_cloture   DECIMAL(18,4)   NOT NULL,
    volume          BIGINT                   DEFAULT 0,
    rendement_jour  DECIMAL(10,6),
    id_instrument   INT             NOT NULL,

    CONSTRAINT pk_prix             PRIMARY KEY (id_prix),
    CONSTRAINT uq_prix_ticker_date UNIQUE (id_instrument, date_cotation),
    CONSTRAINT fk_prix_instrum     FOREIGN KEY (id_instrument)
                                   REFERENCES Instrument(id_instrument)
                                   ON DELETE CASCADE
                                   ON UPDATE CASCADE,
    CONSTRAINT chk_prix_cloture    CHECK (cours_cloture > 0),
    CONSTRAINT chk_prix_volume     CHECK (volume >= 0)
);
```

Figure 7: Creation of the `PrixHistorique` table

```sql
INSERT INTO PrixHistorique (date_cotation, cours_cloture, volume, rendement_jour, id_instrument) VALUES
-- ---- MC.PA (id=1) ----
('2024-08-01', 685.50, 412000, 0.0021, 1),('2024-08-02', 679.20, 398000, -0.0092, 1),
('2024-08-05', 672.80, 521000, -0.0094, 1),('2024-08-06', 681.40, 445000,  0.0128, 1),
('2024-08-07', 688.90, 389000,  0.0110, 1),('2024-08-08', 695.30, 467000,  0.0093, 1),
('2024-08-09', 692.10, 356000, -0.0046, 1),('2024-09-02', 701.50, 478000,  0.0134, 1),
('2024-09-03', 698.70, 412000, -0.0040, 1),('2024-09-04', 705.20, 391000,  0.0093, 1),
('2024-09-05', 711.80, 523000,  0.0094, 1),('2024-09-06', 708.40, 445000, -0.0048, 1),
('2024-10-01', 715.60, 489000,  0.0101, 1),('2024-10-02', 722.30, 412000,  0.0094, 1),
('2024-10-03', 718.90, 367000, -0.0047, 1),('2024-10-04', 725.40, 498000,  0.0090, 1),
('2024-11-04', 731.20, 502000,  0.0080, 1),('2024-11-05', 728.60, 423000, -0.0036, 1),
('2024-11-06', 735.90, 467000,  0.0100, 1),('2024-12-02', 742.10, 534000,  0.0084, 1),
('2024-12-03', 738.50, 489000, -0.0049, 1),('2024-12-04', 745.80, 512000,  0.0099, 1),
-- ---- TTE.PA (id=2) ----
```

```sql
-- ============================================================
--  7. Transaction
-- ============================================================
CREATE TABLE Transaction (
    id_transaction  INT                 NOT NULL AUTO_INCREMENT,
    sens            ENUM('ACHAT','VENTE') NOT NULL,
    quantite        DECIMAL(18,6)       NOT NULL,
    prix_execution  DECIMAL(18,4)       NOT NULL,
    date_transaction DATE               NOT NULL,
    frais           DECIMAL(18,2)       NOT NULL DEFAULT 0,
    id_portefeuille INT                 NOT NULL,
    id_instrument   INT                 NOT NULL,

    CONSTRAINT pk_transaction      PRIMARY KEY (id_transaction),
    CONSTRAINT fk_txn_portef       FOREIGN KEY (id_portefeuille)
                                   REFERENCES Portefeuille(id_portefeuille)
                                   ON DELETE RESTRICT
                                   ON UPDATE CASCADE,
    CONSTRAINT fk_txn_instrum      FOREIGN KEY (id_instrument)
                                   REFERENCES Instrument(id_instrument)
                                   ON DELETE RESTRICT
                                   ON UPDATE CASCADE,
    CONSTRAINT chk_txn_quantite    CHECK (quantite > 0),
    CONSTRAINT chk_txn_prix        CHECK (prix_execution > 0),
    CONSTRAINT chk_txn_frais       CHECK (
        frais >= 0 AND frais <= prix_execution * quantite * 0.05
    )
);
```

Figure 8: Creation of the `Transaction` table

```sql
INSERT INTO Position (id_portefeuille, id_instrument, date_valo, quantite, prix_moyen, poids, pnl_latent) VALUES
-- PF1 : Sécurité Fontaine (prudent – obligations + ETF)
(1, 7,  '2024-12-04', 200.000000,  98.50, 0.4261, 1480.00),
(1, 8,  '2024-12-04', 150.000000, 102.10, 0.3346,  142.50),
(1, 10, '2024-12-04', 120.000000,  77.50, 0.2154,  652.80),
-- PF2 : Équilibré Rousseau (60% actions / 40% obli)
(2, 1,  '2024-12-04',  50.000000, 695.00, 0.4519, 2540.00),
(2, 7,  '2024-12-04', 100.000000,  98.80, 0.1201,  044.00),
(2, 10, '2024-12-04', 200.000000,  78.20, 0.2011,  938.00),
(2, 3,  '2024-12-04',  80.000000,  89.00, 0.0907,  364.80),
(2, 11, '2024-12-04',  30.000000, 545.00, 0.2130, 1286.70),
-- PF3 : Agressif Leclerc (actions + dérivés)
(3, 5,  '2024-12-04', 100.000000, 215.00, 0.1797, 2801.00),
(3, 6,  '2024-12-04',  80.000000, 410.00, 0.2629, 2764.80),
(3, 13, '2024-12-04',  20.000000, 7456.00, 0.1169, 8670.00),
(3, 14, '2024-12-04', 200.000000,  22.00, 0.0624, 4024.00),
```

Figure 14: Extract from the `Position` table

```sql
INSERT INTO Transaction (sens, quantite, prix_execution, date_transaction, frais, id_portefeuille, id_instrument) VALUES
-- === PF1 : Fontaine ===
('ACHAT',  200, 98.20,  '2020-02-20',  9.82,  1, 7),
('ACHAT',  150, 101.80, '2020-03-15',  9.16,  1, 8),
('ACHAT',  120, 77.20,  '2020-04-10',  4.63,  1, 10),
('VENTE',   50, 99.10,  '2024-06-15',  4.96,  1, 7),
('ACHAT',   50, 98.50,  '2024-09-01',  4.93,  1, 7),
-- === PF2 : Rousseau ===
('ACHAT',   50, 672.00, '2019-07-01',  16.80, 2, 1),
('ACHAT',  100, 98.60,  '2019-07-05',  4.93,  2, 7),
```

Figure 15: Extract from the `Transaction` table

4.3 SQL Queries

In an asset management context, a database does not serve merely as a transaction storage system, but acts as an analytical and decision-support tool. The data must enable reconstruction of portfolio positions, performance measurement, exposure identification across asset classes, and detection of certain financial risks.

The SQL queries developed in this section address several business needs: transaction monitoring, client portfolio analysis, financial indicator computation, risk concentration identification, and extraction of statistics useful for business oversight. The objective is to demonstrate how a relational database can be leveraged in a manner consistent with systems used in quantitative finance and asset management.

R1 — Alphabetical List of Financial Instruments

As a quantitative analyst, it is necessary to have a complete and ordered reference of all instruments available in the database, in order to facilitate search and consultation of the asset catalogue.

SQL technique used: Simple `SELECT` with alphabetical `ORDER BY` on the instrument name.

```
SELECT
    id_instrument,
    ticker,
    nom,
    type_instrument,
    secteur,
    devise
FROM Instrument
ORDER BY nom ASC;
```

Result (see Fig 16)

id_instrument	ticker	nom	type_instrument	secteur	devise
4	AIR.PA	Airbus SE	action	Aéronautique	EUR
10	CAC40	Amundi ETF CAC 40	ETF	Indice FR	EUR
5	AAPL	Apple Inc.	action	Technologie	USD
8	BUND10	Bund Allemagne 10 ans 2.5%	obligation	Souverain	EUR
14	AAPL-C	Call AAPL Strike 200 Jan25	dérivé	Technologie	USD
13	CAC-FUT	Future CAC 40 Déc. 2024	dérivé	Indice FR	EUR
11	SP500	iShares Core S&P 500 ETF	ETF	Indice US	USD
1	MC.PA	LVMH Moët Hennessy	action	Luxe	EUR
6	MSFT	Microsoft Corporation	action	Technologie	USD
7	OAT10	OAT France 10 ans 3.0%	obligation	Souverain	EUR
9	CORP-SAN	Obligation Sanofi 5 ans 2.8%	obligation	Santé	EUR
15	EUR-PUT	Put EUR/USD Strike 1.05 Ma...	dérivé	Forex	USD
3	SAN.PA	Sanofi SA	action	Santé	EUR
2	TTE.PA	TotalEnergies SE	action	Énergie	EUR
12	MSCI-EM	Xtrackers MSCI Emerging	ETF	Marchés ém...	USD
NULL	NULL	NULL	NULL	NULL	NULL

Figure 16: Result of R01 — Alphabetical list of instruments

R2 — High Assets Under Management (AUM) Clients

The commercial department wishes to identify high-potential clients in order to offer them advanced investment strategies. Clients whose assets under management (AUM) exceed 100 000 € are selected, which is the threshold characteristic of *aggressive* profiles according to business rule RM04.
SQL technique used: `WHERE` clause with a numeric filter on a `DECIMAL` column, combined with a descending `ORDER BY`.

```
SELECT
    id_client,
    nom,
    prenom,
    email,
    aum,
    profil_risque
FROM Client
WHERE aum > 100000
ORDER BY aum DESC;
```

Result (see Fig 17)

id_client	nom	prenom	email	aum	profil_risque
5	Chevalier	François	f.chevalier@gmail.com	200000.00	agressif
7	Simon	Laurent	l.simon@gmail.com	150000.00	dynamique
3	Leclerc	Antoine	a.leclerc@gmail.com	120000.00	agressif
8	Laurent	Émilie	e.laurent@gmail.com	110000.00	agressif
NULL	NULL	NULL	NULL	NULL	NULL

Figure 17: Result of R02 — Clients with AUM > 100 000 €

R3 — All Transactions for a Given Portfolio

A portfolio manager wishes to consult the complete history of buy and sell operations carried out on a specific portfolio, identified by its `id_portefeuille`, in order to reconstruct the evolution of positions and audit execution quality.
SQL technique used: `WHERE` filter on a parameter-passed identifier, with chronological `ORDER BY`.

```
SELECT
    id_transaction,
    sens,
    quantite,
    prix_execution,
    date_transaction,
    frais,
    id_instrument
FROM Transaction
WHERE id_portefeuille = 3       -- parameter: target portfolio id
ORDER BY date_transaction ASC;
```

Result (see Fig 18)

id_transaction	sens	quantite	prix_execution	date_transaction	frais	id_instrument
15	ACHAT	100.000000	209.8200	2021-01-15	10.49	5
16	ACHAT	80.000000	404.2600	2021-01-15	16.17	6
17	ACHAT	20.000000	7345.7500	2021-02-01	73.46	13
18	ACHAT	200.000000	18.9200	2021-03-10	1.89	14
19	ACHAT	150.000000	144.5000	2021-04-01	7.23	4
20	ACHAT	40.000000	688.9000	2021-05-15	13.78	1
22	ACHAT	50.000000	215.4900	2024-08-02	5.39	5
23	VENTE	100.000000	18.9200	2024-08-05	0.95	14
21	VENTE	50.000000	224.5300	2024-08-08	5.61	5
24	ACHAT	100.000000	22.4500	2024-09-02	1.12	14
25	VENTE	20.000000	7589.5000	2024-09-02	75.90	13
26	ACHAT	20.000000	7645.2500	2024-10-01	76.45	13
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Figure 18: Result of R03 — Transactions for portfolio 3

R4 — Enriched Positions with Instrument Information (INNER JOIN)

The risk manager needs to visualize open positions as of the valuation date of December 4, 2024, enriched with each instrument's characteristics (ticker, type, sector) in order to analyze exposures by asset class and economic sector.
SQL technique used: `INNER JOIN` between `Position` and `Instrument` — only rows with a match in both tables are returned.

```
SELECT
    po.id_portefeuille,
    i.ticker,
    i.nom                   AS instrument,
    i.type_instrument,
    i.secteur,
    po.quantite,
    po.prix_moyen,
    po.poids,
    po.pnl_latent
FROM Position po
INNER JOIN Instrument i ON po.id_instrument = i.id_instrument
WHERE po.date_valo = '2024-12-04'
ORDER BY po.id_portefeuille, po.poids DESC;
```

Result (see Fig 19)

id_portefeuille	ticker	instrument	type_instrument	secteur	quantite	prix_moyen	poids	pnl_latent
1	OAT10	OAT France 10 ans 3.0%	obligation	Souverain	200.000000	98.5000	0.4261	1480.00
1	BUND10	Bund Allemagne 10 ans 2.5%	obligation	Souverain	150.000000	102.1000	0.3346	142.50
1	CAC40	Amundi ETF CAC 40	ETF	Indice FR	120.000000	77.5000	0.2154	652.80
2	MC.PA	LVMH Moët Hennessy	action	Luxe	50.000000	695.0000	0.4519	2540.00
2	SP500	iShares Core S&P 500 ETF	ETF	Indice US	30.000000	545.0000	0.2130	1286.70
2	CAC40	Amundi ETF CAC 40	ETF	Indice FR	200.000000	78.2000	0.2011	938.00
2	OAT10	OAT France 10 ans 3.0%	obligation	Souverain	100.000000	98.8000	0.1201	44.00
2	SAN.PA	Sanofi SA	action	Santé	80.000000	89.0000	0.0907	364.80
3	MSFT	Microsoft Corporation	action	Technologie	80.000000	410.0000	0.2629	2764.80
3	MC.PA	LVMH Moët Hennessy	action	Luxe	40.000000	700.0000	0.2210	1832.00
3	AAPL	Apple Inc.	action	Technologie	100.000000	215.0000	0.1797	2801.00
3	AIR.PA	Airbus SE	action	Aéronautique	150.000000	147.0000	0.1756	1717.50
3	CAC-FUT	Future CAC 40 Déc. 2024	dérivé	Indice FR	20.000000	7456.0000	0.1169	8670.00
3	AAPL-C	Call AAPL Strike 200 Jan25	dérivé	Technologie	200.000000	22.0000	0.0624	4024.00
4	AIR.PA	Airbus SE	action	Aéronautique	200.000000	148.0000	0.3138	2090.00
4	CAC40	Amundi ETF CAC 40	ETF	Indice FR	300.000000	78.0000	0.2464	1467.00
4	TTE.PA	TotalEnergies SE	action	Énergie	400.000000	58.5000	0.2460	1420.00
4	MC.PA	LVMH Moët Hennessy	action	Luxe	60.000000	688.0000	0.0939	3468.00
4	SAN.PA	Sanofi SA	action	Santé	100.000000	90.0000	0.0927	356.00
5	SP500	iShares Core S&P 500 ETF	ETF	Indice US	150.000000	542.0000	0.4102	6883.50
5	AAPL	Apple Inc.	action	Technologie	200.000000	218.0000	0.2261	5002.00
5	MSFT	Microsoft Corporation	action	Technologie	100.000000	418.0000	0.2067	2656.00
5	AAPL-C	Call AAPL Strike 200 Jan25	dérivé	Technologie	100.000000	24.0000	0.0195	1812.00
5	EUR-PUT	Put EUR/USD Strike 1.05 Ma...	dérivé	Forex	500.000000	3.5000	0.0092	365.00
6	OAT10	OAT France 10 ans 3.0%	obligation	Souverain	150.000000	98.4000	0.4159	126.00

Result 1 ×

Figure 19: Result of R04 — Enriched positions as of 2024-12-04

R5 — All Instruments, Including Those with No Transactions (LEFT JOIN)

The reference data management department wishes to identify financial instruments registered in the database that have not yet been involved in any transaction. These "dormant" instruments may signal obsolete data or assets not yet allocated.
SQL technique used: `LEFT JOIN` between `Instrument` and `Transaction` — all instruments appear; the `id_transaction` column is `NULL` for those with no transaction.

```
SELECT
    i.id_instrument,
    i.ticker,
    i.nom,
    i.type_instrument,
    t.id_transaction        -- NULL if no transaction
FROM Instrument i
```

```
LEFT JOIN Transaction t ON i.id_instrument = t.id_instrument
ORDER BY t.id_transaction IS NULL DESC, i.ticker ASC;
```

Result (see Fig 20)

id_instrument	ticker	nom	type_instrument	id_transaction
5	AAPL	Apple Inc.	action	15
5	AAPL	Apple Inc.	action	21
5	AAPL	Apple Inc.	action	22
5	AAPL	Apple Inc.	action	34
5	AAPL	Apple Inc.	action	39
5	AAPL	Apple Inc.	action	40
5	AAPL	Apple Inc.	action	44
5	AAPL	Apple Inc.	action	48
5	AAPL	Apple Inc.	action	49
5	AAPL	Apple Inc.	action	50
14	AAP...	Call AAPL...	dérivé	52
14	AAP...	Call AAPL...	dérivé	38
14	AAP...	Call AAPL...	dérivé	24
14	AAP...	Call AAPL...	dérivé	18
14	AAP...	Call AAPL...	dérivé	23
14	AAP...	Call AAPL...	dérivé	54
14	AAP...	Call AAPL...	dérivé	55
4	AIR.PA	Airbus SE	action	19
4	AIR.PA	Airbus SE	action	29
8	BUN...	Bund Alle...	obligation	2
8	BUN...	Bund Alle...	obligation	42
13	CAC...	Future C...	dérivé	26
13	CAC...	Future C...	dérivé	25
13	CAC...	Future C...	dérivé	17
10	CAC40	Amundi E...	ETF	47

Result 1

Figure 20: Result of R05 — Instruments with or without transactions

R6 — Total Invested Value by Instrument Type

The investment committee wishes to know the distribution of invested capital by asset class across all buy transactions, in order to manage the overall strategic allocation. The gross invested amount is computed as `quantity * execution_price`.

SQL technique used: `SUM` aggregation with `GROUP BY` on a join between `Transaction` and `Instrument`.

```
SELECT
    i.type_instrument,
    COUNT(t.id_transaction)                         AS nb_transactions,
```

```
    ROUND(SUM(t.quantite * t.prix_execution), 2)    AS total_invested_amount,
    ROUND(AVG(t.prix_execution), 4)                  AS avg_execution_price
FROM Transaction t
INNER JOIN Instrument i ON t.id_instrument = i.id_instrument
WHERE t.sens = 'ACHAT'
GROUP BY i.type_instrument
ORDER BY total_invested_amount DESC;
```

Result (see Fig 21)

type_instrument	nb_transactions	montant_total_investi	prix_moyen_execution
action	21	549284.80	336.6157
dérivé	8	317930.00	1887.6925
ETF	8	216196.50	238.0538
obligation	8	89592.00	99.5400

Figure 21: Result of R06 — Total invested amount by instrument type

R7 — Number of Positions per Portfolio (Descending)

The reporting department wishes to produce a summary dashboard showing the number of open position lines per portfolio at the latest valuation date, in order to assess the diversification level of each mandate.

SQL technique used: `COUNT` with `GROUP BY` on `id_portefeuille`, sorted in descending order.

```
SELECT
    po.id_portefeuille,
    p.nom                     AS portfolio_name,
    COUNT(*)                  AS nb_positions,
    ROUND(SUM(po.poids), 4)   AS total_weight
FROM Position po
INNER JOIN Portefeuille p ON po.id_portefeuille = p.id_portefeuille
WHERE po.date_valo = '2024-12-04'
GROUP BY po.id_portefeuille, p.nom
ORDER BY nb_positions DESC;
```

Result (see Fig 22)

id_portefeuille	nom_portefeuille	nb_positions	somme_poids
3	PF Agressif Leclerc	6	1.0185
2	PF Équilibré Rousseau	5	1.0768
4	PF Dynamique Garnier	5	0.9928
5	PF Global Chevalier	5	0.8717
7	PF Croissance Simon	4	0.9129
8	PF Tech Laurent	4	0.7270
1	PF Sécurité Fontaine	3	0.9761
6	PF Prudent Morel	3	0.9846

Figure 22: Result of R07 — Number of positions per portfolio

R8 — Managers Overseeing More Than Two Clients

Management wishes to identify portfolio managers with a significant workload, namely those simultaneously overseeing more than two clients, in order to assess recruitment needs and prevent operational risks associated with overload.
SQL technique used: `HAVING` with `COUNT` to filter groups after aggregation.

```
SELECT
    g.id_gestionnaire,
    g.nom,
    g.prenom,
    g.specialite,
    COUNT(c.id_client) AS nb_clients
FROM Gestionnaire g
INNER JOIN Client c ON g.id_gestionnaire = c.id_gestionnaire
GROUP BY g.id_gestionnaire, g.nom, g.prenom, g.specialite
HAVING COUNT(c.id_client) >= 2
ORDER BY nb_clients DESC;
```

Result (see Fig 23)

id_gestionnaire	nom	prenom	specialite	nb_clients
2	Martin	Sophie	Obligations souveraines	2
3	Bernard	Julien	Dérivés et produits structurés	2

Figure 23: Result of R08 — Managers overseeing more than 2 clients

R9 — Average Latent PnL by Instrument Type (Filtered)

The risk manager wishes to measure the average latent performance by asset class across all portfolios, retaining only those classes generating an average latent *PnL* above 500 €, in order to target the best-performing allocations at the current valuation date.
SQL technique used: `AVG` and `SUM` with `GROUP BY` and post-aggregation filtering via `HAVING`.

```
SELECT
    i.type_instrument,
    COUNT(*)                       AS nb_positions,
    ROUND(AVG(po.pnl_latent), 2)   AS avg_pnl,
    ROUND(SUM(po.pnl_latent), 2)   AS total_pnl
FROM Position po
INNER JOIN Instrument i ON po.id_instrument = i.id_instrument
WHERE po.date_valo = '2024-12-04'
GROUP BY i.type_instrument
HAVING AVG(po.pnl_latent) > 500
ORDER BY avg_pnl DESC;
```

Result (see Fig 24)

type_instrument	nb_positions	pnl_moyen	pnl_total
dérivé	5	4301.40	21507.00
action	16	2489.73	39835.60
ETF	8	2109.13	16873.00

Figure 24: Result of R09 — Average latent PnL by instrument type > 500 €

R10 — Maximum and Minimum Closing Price per Instrument

The quantitative research department needs to know the historical price extrema for each instrument over the observation period (August–December 2024), in order to compute price variation ranges and calibrate Value-at-Risk models.
SQL technique used: `MAX` and `MIN` with `GROUP BY` on `id_instrument`, enriched with a `JOIN` to display the ticker.

```
SELECT
    i.ticker,
    i.nom,
    i.type_instrument,
    MIN(ph.cours_cloture)                              AS min_price,
    MAX(ph.cours_cloture)                              AS max_price,
    ROUND(MAX(ph.cours_cloture)
          - MIN(ph.cours_cloture), 4)                  AS price_range,
    ROUND((MAX(ph.cours_cloture)
           - MIN(ph.cours_cloture))
          / MIN(ph.cours_cloture) * 100, 2)            AS range_pct
FROM PrixHistorique ph
INNER JOIN Instrument i ON ph.id_instrument = i.id_instrument
GROUP BY i.id_instrument, i.ticker, i.nom, i.type_instrument
ORDER BY range_pct DESC;
```

Result (see Fig 25)

ticker	nom	type_instrument	cours_min	cours_max	amplitude	amplitude_pct
AAPL-C	Call AAPL Strike 200 Jan25	dérivé	18.9200	42.1200	23.2000	122.62
EUR-PUT	Put EUR/USD Strike 1.05 Mar25	dérivé	3.2300	4.5600	1.3300	41.18
AAPL	Apple Inc.	action	209.8200	243.0100	33.1900	15.82
SP500	iShares Core S&P 500 ETF	ETF	518.4500	587.8900	69.4400	13.39
MC.PA	LVMH Moët Hennessy	action	672.8000	745.8000	73.0000	10.85
MSFT	Microsoft Corporation	action	404.2600	444.5600	40.3000	9.97
AIR.PA	Airbus SE	action	144.5000	158.4500	13.9500	9.65
TTE.PA	TotalEnergies SE	action	57.1200	62.0500	4.9300	8.63
CAC40	Amundi ETF CAC 40	ETF	76.5400	82.8900	6.3500	8.30
CAC-FUT	Future CAC 40 Déc. 2024	dérivé	7345.7500	7889.5000	543.7500	7.40
MSCI-EM	Xtrackers MSCI Emerging	ETF	40.2300	43.1200	2.8900	7.18
SAN.PA	Sanofi SA	action	87.4500	93.5600	6.1100	6.99
OAT10	OAT France 10 ans 3.0%	obligation	98.3800	99.2400	0.8600	0.87
BUND10	Bund Allemagne 10 ans 2.5%	obligation	102.2800	103.0500	0.7700	0.75
CORP-SAN	Obligation Sanofi 5 ans 2.8%	obligation	100.0800	100.4200	0.3400	0.34

Figure 25: Result of R10 — Price extrema per instrument

R11 — Portfolios Whose Net Asset Value Exceeds the Average (Scalar Subquery)

The performance committee wishes to identify portfolios whose net asset value exceeds the average across all managed mandates, in order to highlight outperforming strategies and analyze the factors driving their success.
SQL technique used: Scalar subquery in the `WHERE` clause returning the global average via `AVG`, compared against each row of the main query.

```
SELECT
    id_portefeuille,
    nom,
    devise_base,
    strategie,
    valeur_liquidative,
    ROUND(valeur_liquidative - (
        SELECT AVG(valeur_liquidative) FROM Portefeuille
    ), 2)  AS deviation_from_average
FROM Portefeuille
WHERE valeur_liquidative > (
    SELECT AVG(valeur_liquidative)
    FROM   Portefeuille
)
ORDER BY valeur_liquidative DESC;
```

Result (see Fig 26)

id_portefeuille	nom	devise_base	strategie	valeur_liquidative	ecart_a_la_moyenne
5	PF Global Chevalier	USD	Global macro multi-asset	215000.00	103062.50
7	PF Croissance Simon	EUR	Actions tech + ETF sectoriels	162000.00	50062.50
3	PF Agressif Leclerc	EUR	Actions croissance + dérivés	135000.00	23062.50
8	PF Tech Laurent	USD	Actions tech US + dérivés	118000.00	6062.50

Figure 26: Result of R11 — Portfolios above the average NAV

R12 — Instruments with Only Buy Transactions (NOT EXISTS)

The compliance department wishes to identify instruments on which no sale has ever been recorded in the database. These assets are held in a strictly long position (*buy-and-hold*), which may indicate either a deliberate passive strategy or a lack of liquidity on these securities.
SQL technique used: `NOT EXISTS` with a correlated subquery, which verifies for each instrument the absence of any transaction of type `SELL`.

```
SELECT
    i.id_instrument,
    i.ticker,
    i.nom,
    i.type_instrument,
    i.secteur
FROM Instrument i
WHERE EXISTS (
    -- the instrument has at least one BUY
    SELECT 1
    FROM   Transaction t
    WHERE  t.id_instrument = i.id_instrument
```

```
    AND    t.sens = 'ACHAT'
)
AND NOT EXISTS (
    -- but no SELL
    SELECT 1
    FROM   Transaction t
    WHERE  t.id_instrument = i.id_instrument
    AND    t.sens = 'VENTE'
)
ORDER BY i.type_instrument, i.ticker;
```

Result (see Fig 27)

id_instrument	ticker	nom	type_instrument	secteur
4	AIR.PA	Airbus SE	action	Aéronautique
6	MSFT	Microsoft Corporation	action	Technologie
3	SAN.PA	Sanofi SA	action	Santé
2	TTE.PA	TotalEnergies SE	action	Énergie
8	BUND10	Bund Allemagne 10 ans 2.5%	obligation	Souverain
9	CORP-SAN	Obligation Sanofi 5 ans 2.8%	obligation	Santé
10	CAC40	Amundi ETF CAC 40	ETF	Indice FR
12	MSCI-EM	Xtrackers MSCI Emerging	ETF	Marchés émerg.
11	SP500	iShares Core S&P 500 ETF	ETF	Indice US
15	EUR-PUT	Put EUR/USD Strike 1.05 Mar25	dérivé	Forex
NULL	NULL	NULL	NULL	NULL

Figure 27: Result of R12 — Instruments with no recorded sales

R13 — Portfolio Ranking with Tie-Breaking

The risk management department wishes to evaluate the size and history of portfolios in order to adjust regulatory leverage limits. A rigorous ranking is required: portfolios are ordered by descending net asset value; in the event of a tie, the oldest portfolio takes precedence (chronological tie-breaking).
SQL technique used: Multi-column `ORDER BY` with mixed directions (`DESC` / `ASC`).

```
SELECT
    id_portefeuille,
    nom,
    valeur_liquidative,
    date_creation
FROM Portefeuille
ORDER BY
    valeur_liquidative DESC,   -- primary criterion: descending NAV
    date_creation      ASC;    -- tie-breaker: oldest portfolio first
```

Result (see Fig 28)

	id_portefeuille	nom	valeur_liquidative	date_creation
▶	5	PF Global Chevalier	215000.00	2017-04-01
	7	PF Croissance Simon	162000.00	2020-09-01
	3	PF Agressif Leclerc	135000.00	2021-01-10
	8	PF Tech Laurent	118000.00	2019-12-10
	4	PF Dynamique Garnier	101000.00	2018-11-25
	2	PF Équilibré Rousseau	82500.00	2019-06-20
	1	PF Sécurité Fontaine	46200.00	2020-02-15
	6	PF Prudent Morel	35800.00	2022-05-20
*	NULL	NULL	NULL	NULL

Figure 28: Result of R13 — Portfolio ranking with tie-breaking

R14 — Multi-Asset Diversification Audit of Portfolios

The compliance department wishes to verify that portfolio managers respect the diversification mandates imposed by clients and do not expose funds to excessive idiosyncratic risk. The objective is to identify portfolios that have executed transactions across **at least two distinct financial instrument types** (`equity`, `bond`, `ETF`, `derivative`).
SQL technique used: Subquery with `COUNT(DISTINCT ...)` aggregation in the `HAVING` clause.

```
SELECT
    p.id_portefeuille,
    p.nom
FROM Portefeuille p
WHERE p.id_portefeuille IN (
    SELECT      t.id_portefeuille
    FROM        Transaction  t
    JOIN        Instrument   i ON t.id_instrument = i.id_instrument
    GROUP BY    t.id_portefeuille
    HAVING      COUNT(DISTINCT i.type_instrument) >= 2
)
ORDER BY p.id_portefeuille;
```

Result (see Fig 29)

R15 — Most Volatile Assets by Asset Class (Local Maximum)

As a quantitative researcher, it is necessary to map extreme market risks (*tail risk*) in order to calibrate hedging models and option pricing. For each asset class (`type_instrument`), the instrument exhibiting the highest volatility is identified. In the event of a tie within the same category, **all tied instruments are displayed**.
SQL technique used: Join with a scalar subquery returning the local maximum per group (`MAX` with `GROUP BY`), which naturally handles ties.

```
SELECT
    i.type_instrument,
    i.ticker,
    i.nom,
    i.volatilite
FROM Instrument i
```

	id_portefeuille	nom
▶	1	PF Sécurité Fontaine
	2	PF Équilibré Rousseau
	3	PF Agressif Leclerc
	4	PF Dynamique Garnier
	5	PF Global Chevalier
	7	PF Croissance Simon
	8	PF Tech Laurent
*	NULL	NULL

Figure 29: Result of R14 — Portfolios with multi-asset diversification

```
JOIN (
    SELECT
        type_instrument,
        MAX(volatilite) AS vol_max
    FROM  Instrument
    GROUP BY type_instrument
) max_vol
    ON  i.type_instrument = max_vol.type_instrument
    AND i.volatilite      = max_vol.vol_max
ORDER BY
    i.type_instrument ASC,
    i.ticker          ASC;
```

Result (see Fig 30)

	type_instrument	ticker	nom	volatilite
▶	action	AAPL	Apple Inc.	0.2632
	obligation	CORP-SAN	Obligation Sanofi 5 ans 2.8%	0.0521
	ETF	MSCI-EM	Xtrackers MSCI Emerging	0.2210
	dérivé	AAPL-C	Call AAPL Strike 200 Jan25	0.4512

Figure 30: Result of R15 — Most volatile instrument per asset class

5 Conclusion

This project allowed us to apply the core concepts of relational database design and management in a complex and realistic domain: financial portfolio management. We designed a coherent data

model, implemented a MySQL database with integrity constraints, developed advanced SQL queries, and built a console application to interact with the data.
This work also deepened our understanding of the challenges related to data structuring, consistency, and exploitation in a context close to real-world information systems.